

1. Administrative & Study Identification

Full Study Title (Official Title): An Open-label Extension Study to Evaluate Safety and Tolerability of Sotatercept (MK-7962) Administered Using a Weight-banded Approach in Participants With Pulmonary Arterial Hypertension (PAH) on Standard of Care **Brief Title:** A Clinical Study of Sotatercept (MK-7962) in People With Pulmonary Arterial Hypertension (MK-7962-031/LIGHTRAY EXTENSION) **Short Title/Acronym:** LIGHTRAY EXTENSION **Protocol Number:** MK-7962-031 **NCT Number:** NCT06925750 **Trial Status:** Recruiting **Sponsor/Company Name:** Merck **Investigational Product:** Sotatercept (MK-7962) **Phase of Development:** Phase II **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of weight-banded doses of sotatercept administered over a longer period of time in participants with PAH. **Secondary Objectives:** To assess the long-term effects of sotatercept on cardiopulmonary performance, disease progression, and maintenance of symptom improvements.

2.2 Background & Rationale Sotatercept is a novel therapy designed to treat the underlying disease process of pulmonary arterial hypertension (PAH) by acting as an activin signaling inhibitor. A past study, MK-7962-024 (LIGHTRAY) (NCT06664801), learned about the safety and effects of sotatercept in people with PAH. One of the goals of that study was to learn about sotatercept when given at a dose based on the weight range a person is in (weight-banded doses) compared to when given based on a person's exact weight. This open-label extension study (LIGHTRAY EXTENSION) aims to gather long-term safety and tolerability data for participants who successfully completed the initial LIGHTRAY trial, ensuring continued therapeutic access and observation of prolonged cardiopulmonary benefits.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Extension Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Rollover extension cohort)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 130 (Rollover participants from the MK-7962-024 (LIGHTRAY) parent study) **Number of Sites:** Multicenter (Global, including sites in

Australia, Canada, etc.) **Estimated Study Start:** 24-Jul-2025
(Registration Date) **Duration:** Up to 24 months of treatment per participant

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Has completed the treatment period of study MK-7962-024 (LIGHTRAY) (including Visit 11) on study intervention. 3. Did not discontinue study intervention in the parent trial and is able to safely enroll into MK-7962-031 (LIGHTRAY EXTENSION). 4. Has not started treatment with commercially available sotatercept.

4.2 Exclusion Criteria 1. Has current exposure or is planning to begin treatment with an activin signaling inhibitor (other than sotatercept). 2. Any condition, comorbidity, or severe adverse event from the parent study that would preclude safe ongoing participation in the extension study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sotatercept administered via subcutaneous (SC) injection every 3 weeks. Dosing utilizes a weight-banded method, with an initial dose of up to 45 mg progressing to a maintenance dose of up to 90 mg. All eligible participants will receive dosing at the level (initial or maintenance) at which they finished the parent LIGHTRAY study. **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** Up to 24 months

5.2 Concomitant & Prohibited Therapy Permitted: Participants will continue to take their background standard-of-care PAH therapy during the study. **Prohibited:** Commercially available sotatercept (outside of the trial) or any other investigational activin signaling inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening/Rollover	Day 0 Informed Consent, verification of LIGHTRAY completion, background therapy review
Treatment Visits	Every 3 weeks	Sotatercept SC injection (weight-banded dose), Safety & AE monitoring, vital signs	
Interim Assessments	Periodic	Cardiopulmonary evaluation, exercise capacity tracking, quality of life questionnaires	
End of Study	Month 24	Final clinical assessment, end-of-treatment safety labs, exit interview	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Long-term safety and tolerability assessed by: 1. Number of Participants Who Experience One or More Adverse Events (AEs): An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. 2. Number of Participants who Discontinue Study Intervention Due to an AE: The number of participants who discontinue study intervention due to an AE will be reported.

7.2 Secondary & Exploratory Endpoints Long-term cardiopulmonary performance and exercise capacity (e.g., maintenance of improvements achieved in the parent study). Overall quality of life and progression of pulmonary arterial hypertension (PAH) symptoms.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the extension study. An AE is defined as any untoward medical occurrence temporally associated with the use of the study intervention, whether or not considered related to the drug.

Serious Adverse Events (SAEs): Must be reported to the sponsor and regulatory authorities within standard rapid reporting timelines (e.g., 24 hours). **Data Monitoring Committee (DMC):** Independent safety committee monitoring long-term safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of sotatercept in the extension phase). **Sample Size Justification:** Target enrollment of 130 is based on the number of eligible patients expected to rollover from the MK-7962-024 (LIGHTRAY) parent study. **Handling of Missing Data:** Summarized descriptively; discontinuations due to AEs will be explicitly tracked and analyzed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical site monitoring and centralized data review to ensure adherence to the weight-banded dosing protocol and accurate AE reporting. **Audit Trail:** Electronic case report forms (eCRF) will be utilized for accurate capture of rollover data and ongoing long-term follow-up metrics.

11. Study Identifiers & Governance

Primary ID: MK-7962-031 **Registry Number:** NCT06925750 **Sponsor/Lead Organization:** Merck **Study Title:** LIGHTRAY EXTENSION **Official Regulatory Title:** An Open-label Extension Study to Evaluate Safety and Tolerability of Sotatercept (MK-7962) Administered Using a Weight-banded Approach in Participants With Pulmonary Arterial Hypertension (PAH) on Standard of Care

12. Clinical Framework (PAH Specific)

Disease Area / Condition: Pulmonary Arterial Hypertension (PAH) **Diagnosis Threshold:** Clinically confirmed PAH requiring background standard-of-care therapy. **Medical Methodology:** Activin signaling inhibition. **Optimization Target:** Sustained improvement in cardiopulmonary function and symptom reduction (WHO functional class).

13. Study Procedures & Methodology

Management Pathway: Integration of sotatercept injections with ongoing background PAH therapies. **Education Protocol (HCP):** Site training on the protocol-specific weight-banded dosing calculations and SC administration. **Education Protocol (Patient):** Counseling on long-term expectations of the open-label extension phase. **Quality Audit Mechanism:** Continuous safety data reviews by independent committees to evaluate the long-term safety profile.

14. Observation & Follow-Up Schedule

Total Duration: Up to 24 months **Onsite Visit Cadence:** Every 3 weeks for drug administration and ongoing safety checks. **Remote Monitoring Frequency:** N/A (Predominantly onsite due to SC injection requirements) **Checklist Review Interval:** Continuous AE reporting at each 3-week visit cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				